**YOU CAN SIMULATE USING PROTEUS SOFTWARE, TINKERCAD CIRCUITS (ONLINE) AND WOKWI (ONLINE)**

1) Controlling actuators through Serial Monitor. Creating different LED patterns and controlling them using push button switches. Controlling servo motor with the help of joystick.
2) Calculate the distance to an object with the help of an ultrasonic sensor and display it on an LCD
3) (a) Controlling relay state based on ambient light levels using LDR sensor.
   (b) Basic Burglar alarm security system with the help of PIR sensor and buzzer.
   (c) Displaying humidity and temperature values on LCD
4) (a) Controlling relay state based on input from IR sensors.
   (b) Interfacing stepper motor with R-Pi.
   (c) Advanced burglar alarm security system with the help of PIR sensor, buzzer and keypad. (Alarm gets disabled if correct keypad password is entered)
   (d) Automated LED light control based on input from PIR (to detect if people are present) and LDR (ambient light level).
5) Upload humidity & temperature data to Thing Speak, periodically logging ambient light level to Thing Speak.
6) Controlling LEDs, relay & buzzer using Blynk app.
7) Introduction to HTTP. Hosting a basic server from the ESP32 to control various digital based actuators (led, buzzer, relay) from a simple web page.
8) Displaying various sensor readings on a simple web page hosted on the ESP32.
9) Controlling LEDs/Motors from an Android/Web app, Controlling AC Appliances from an android/web app with the help of relay.
10) Displaying humidity and temperature data on a web-based application.